Motor Imagery EEG Signal Classification based on Deep Transfer Learning

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***Abstract*—Deep transfer learning (DTL) has developed rapidly in the field of motor imagery (MI) on brain-computer interface (BCI) in recent years. DTL utilizes deep neural networks with strong generalization capabilities as the pre-training framework and automatically extracts richer and more expressive features during the training process. The goal of this paper is utilizing the DTL to classify MI electroencephalogram (EEG) signals on the premise of a small data set. The publicly available dataset III of the second BCI competition is applied in both the training part and testing part to evaluate the effectiveness of the proposed method. Firstly in the process, finite impulse response (FIR) filter and wavelet transform threshold denoising method are used to remove redundant signals and artifacts in EEG signals. Then, the continuous wavelet transform (CWT) is utilized to convert the one-dimensional EEG signal into a two-dimensional time-frequency amplitude representation as the input of the pre-trained convolutional neural network (CNN) for classifying two types of MI signals. Employing the input data of 140 trials for training, the final classification accuracy rate reaches 96.43%. Compared with the results of some superior machine learning models using the same data set, the accuracy and Kappa value of this DTL model are better. Therefore, the proposed scheme of MI EEG signal classification based on the DTL method offers preferably empirical performance.**

***Index Terms*—Brain-Computer Interface, Deep Transfer Learning, Convolutional Neural Network, Motor Imagery.**

I. Introduction

With the popular applications in smart auxiliary robots and rehabilitation apparatus in recent years, the brain-computer interface (BCI) as the significant underlying technology has become a key research object. BCI provides a direct channel between the human brain and the computer or other external devices, by collecting biological signals related to the human brain's neural activity, and analyze and convert them into commands to control the external devices such as rehabilitation robots. Electroencephalogram (EEG) owns the feature of the collection of brain signals directly from the outside of the brain without surgical implantation of the gray matter or cranial cavity, which would avoid triggering the immune response and subsequent trauma. The EEG signals collected by the device are based on data with descriptive values such as frequency band and amplitude. Among these waves classified by the frequency domain, Mu and Beta waves have significant connections with motor imagery (MI) signals, because the changes in their waveforms largely depend on the cortex of the human brain which is responsible for normal movement. When the subject is in a stationary state, the rhythm of Mu and Beta reaches the highest power value. When the subjects try to move or imagine moving their body parts, the power of their rhythm changes accordingly and begins to produce an intermittent downward fluctuation. Because of the MI on different body parts such as hands, legs and feet, the rhythm changes are also correspondingly different. By performing the signal processing, feature extraction, and model training on the collected brainwave signals, an accurate judgment can be made on the MI activities of the subject.

There are numerous examples of traditional mainstream machine learning methods applied to the BCI classification of MI. Bhardwaj et al. uses linear discriminant analysis (LDA) to linearly combine the extracted features to separate the real human emotions. However, the model does not apply to nonlinear situations. Then Bhardwaj et al. adopted another model, support vector machine (SVM) to classify different categories by looking for nonlinear boundaries avoiding the defect of incompatible nonlinearity, and improving the final accuracy [1]. Moreover, manifold learning such as Riemannian geometry is also widely applied as the MI classification model and obtains better mean Kappa value for accuracy [2]. Although traditional machine learning has achieved very excellent classification results, deep learning (DL) is gaining popularity and becoming the next mainstream approach with the advantages of utilizing big data and plentiful computing resources to have pushed the boundaries of what was possible [3]. The concept of deep learning was first proposed in 2006 by Hinton et al., who also verified that the neural network structure with multiple hidden layers has excellent learning ability for the essential characteristics of data [4]. However, the shortcoming of deep learning is also very obvious and severe, that is, it has a strong dependence on massive training data. Deep neural networks need to learn the underlying essential characteristics of the data through a large amount of data so that subsequent layers can successfully identify and reach the final decision. Due to the large differences in the EEG signals collected from human subjects, the lack of training data makes it generally difficult to obtain a universal DL model for BCI research. Especially when facing a new session on the same topic, it is also necessary to recollect a large

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number of high-quality, annotated data sets as the training input of the model, but the fact is that such data collection is undoubtedly extremely troublesome, time-consuming, and expensive. Furthermore, each data sample often corresponds to a patient suffering from a painful clinical trial, which leads to the collection for EEG signals facing with the moral and ethical problems.

Yosinski et al. published an article of great significance for the follow-up research of deep transfer learning (DTL) at the top machine learning conference NIPS in 2014 [5]. Yosinski et al. first affirmed the portability of deep neural networks and used actual experiments to prove that deep transfer networks are better than randomly initialized weights, and the transfer of network layers can accelerate model learning and optimization. Finetune, which uses the trained network to adjust training for new tasks, saves a lot of time cost of retraining. The use of pre-trained models will indirectly expand the training data set, making the model more robust and generalized. So far, numerous scholars have begun to apply DTL to various fields (including BCI) to remedy the lack of data in the target domain.

In view of the above discussion, a new MI classification model that combines the continuous wavelet transform (CWT) method and DTL method will be proposed in this paper. This method improves the final classification accuracy and Kappa coefficient by adopting the CWT to convert the one-dimensional EEG signal into a two-dimensional time-frequency amplitude representation as to the input of the pre-trained convolutional neural network (CNN).

The rest of the paper is organized as follows: in Section II, the related methodologies including finite impulse response bandpass filter, wavelet transform threshold denoising, CWT, CNN and DTL are recommended firstly for facilitating understanding the principles. In Section III, the experiment of data preprocessing, feature extraction and network training are demonstrated in detail. In Section IV, the results and analysis compared with the results of the aforementioned non-DTL models on the same data set are discussed. In Section V, conclusion based on the existing achievements is summarized.

II. Methodology

A. Finite Impulse Response Bandpass Filter & Wavelet Transform Threshold Denoising

There are two kinds of artifacts in the recorded EEG signals, which are non-physiological artifacts and physiological artifacts. Among them, non-physiological artifacts are mainly produced by 50 Hz power frequency interference. In this paper, it is eliminated by using the finite impulse response (FIR) bandpass filter. The FIR filter performs filtering by convolution with the filter coefficients and the signal, which can strictly retain the linear phase-frequency characteristics while retaining the amplitude-frequency characteristics of the EEG effective signal. Because the frequency of brain waves in human EEG is generally 0.5-30 Hz, and the μ waves and β waves related to motor imagination mainly exist in 8-13 Hz and 13-30 Hz. Therefore, the band-pass filter is set to only retain the effective EEG band of 8-30 Hz. Physiological artifacts are mainly caused by electrical potential activities outside the brain, including electrooculogram (EOG) artifacts (blinking and eye movement), electromyography (EMG) artifacts, and electrocardiographic (ECG) artifacts. ECG artifacts have relatively little influence on EEG, so most researchers generally ignore them as a simplification. It can also be removed through component separation. The main frequency of EMG artifacts is above 30 Hz, so most of them have been removed by the previous FIR bandpass filter. However, EOG artifacts have high-energy signals in low-frequency bands and the signal amplitude is much larger than EEG signals, which will seriously affect the analysis of EEG signals. Therefore, this will be the focus of the denoising stage in the signal processing of the project. During the rotation of the human eyeball, the direction of the electric dipole between the cornea and the sclera will change relative to the recording electrode. The EOG artifact mainly refers to the potential difference generated by the electric dipole between the initial position and the end position of the eyeball (the electrode with the same direction as the eye movement records positive EOG artifacts, but the opposite direction means the negative EOG artifacts). The wavelet transform threshold denoising method firstly proposed by Donoho et al. in 1992 was adopted to eliminate the EOG artifact [6]. It is a non-linear denoising method that can reach approximately optimal under the minimum mean square error. The overall process for eliminating noised signal is summarized in Figure 1.

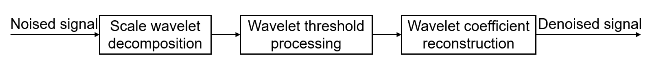

Fig. 1. Process of Wavelet Transform Threshold Denoising Method

B. Continuous Wavelet Transformation

The principle of CWT is to use wavelets of different scales to deconvolve window by window to finally obtain the wavelet coefficient matrix. The equation of CWT is shown in Equation 1.

$$CWT_x^{\psi}(\tau, s) = \frac{1}{\sqrt{s}} \int_{-\infty}^{+\infty} x(t)\psi\left(\frac{t}{s} - \tau\right) dt \qquad (1)$$

- τ is the translation factor (time shift).
- s is the scaling factor (dilation or compression).
- ψ is wavelet base (continuous function in both the time domain and the frequency domain to generate the daughter wavelets which are simply the translated and scaled versions of the mother wavelet).

CWT can transform the input data sequence $x(t)$ into a convolution with a set of functions generated by the mother wavelet ψ. In general, the fast Fourier transform (FFT) algorithm is utilized to compute the results of the convolution [7]. However, the mother wavelets are of diversity and when different mother wavelets are used to analyze the same problem, there may be a huge gap in the possible results. There is no definite method to directly judge, mainly by using the continuous wavelet analysis method to process the signal result and the error of the theoretical result to judge the quality of the wavelet base. This paper adopted the famous Morlet wavelet which has the feature of equal variance in time and frequency. The complex trigonometric function multiplied by the exponential decay function constitutes the Morlet wavelet function (ω_0 is the center frequency) shown in Equation 2.

$$\psi(t) = \exp\left(i\omega_0 t\right)\exp\left(-\frac{t^2}{2}\right) \tag{2}$$

In Equation 2, the complex trigonometric function expresses the volatility of the Morlet wavelet basis, so that CWT can analyze the frequency (and the original signal product integration to find the maximum value). The attenuation function expresses its limited support so that CWT can locate the time. By combining the two, the Morlet wavelet can be used for time-frequency analysis. Finally, the square of output $CWT_x^{\psi}(\tau, s)^2$ can be calculated as the power spectrum, which is also called scalogram representing the energy of the signal computed using the square modulus of the CWT [8]. The relation between scale and frequency is reflected in the axes: the low-frequency signal generates finer fringes under the transformation of CWT, which means higher frequency resolution. But at the same time, the fringe interval will exceed the actual signal interval, which means lower time resolution. The high-frequency signal will be the opposite, with lower frequency resolution and higher time resolution.

C. Convolutional Neural Network

The CNN is an improved version with changing the function and form of the traditional neural network. In the framework of CNN, the two parts of feature extraction (convolution, activation function, pooling) and classification and recognition (fully connected layer) are included. AlexNet is the most representative algorithm in the CNN. It won the 2012 ImageNet competition with accuracy far exceeding other traditional machine learning, shocked the entire computer vision industry, and established the CNN to occupy the mainstream position of machine learning. AlexNet consists of five convolutional layers (the maximum pooling layer is connected after the three convolutional layers) and three fully connected layers. Its innovative features are as follows.

- Adopt unsaturated rectified linear unit (ReLU).
- Propose local response normalization (LRN).
- Employ maximum pooling for all feature maps.
- Apply dropout in the full connection layer.

ReLU is adopted as an activation function to replace the traditional Tanh and sigmoid functions, which speeds up the process of network training, reduces the complexity of calculations, and avoids gradient divergence problems such as Tanh and Logistic activation functions to a certain extent. Next AlexNet innovatively proposes LRN after activation and pooling, creating a competition mechanism for the activity of local neurons, making the value with larger response become relatively larger, and inhibiting other neurons with smaller feedback to enhance the generalization of the model ability. After that, all the feature maps are employed the maximum pooling method, which replaces the previous CNN model that has always used average pooling to avoid the blurring effect brought by average pooling. The step size is set to be smaller than the size of the pooling core to ensure that there will be overlap and coverage between the outputs of the pooling layer to enhance the richness of features. Dropout is applied in the final fully connected layer to randomly ignore a part of the neural network unit to alleviate the over-fitting phenomenon of the neural network and avoid falling into the local optimum when optimizing the network parameters and enhance the generalization ability of the network.

D. Deep Transfer Learning

DTL refers to the use of deep neural networks as tools to carry out the transfer learning, which is about how to use knowledge in other fields for effective knowledge transfer. DTL method solves the problems encountered in deep learning that are difficult to improve due to insufficient training data, and it also greatly saves the time cost of retraining the model. DTL method is firstly supposing that there is a task $\mathcal{T}_t$ based on the target domain $\mathcal{D}_t$, which can be helped from the source domain $\mathcal{D}_s$ of another task $\mathcal{T}_s$ ($\mathcal{D}_t \neq \mathcal{D}_s$ or $\mathcal{T}_t \neq \mathcal{T}_s$). DTL aims to discover and transfer the potentially transferable knowledge of $\mathcal{D}_s$ and $\mathcal{T}_s$ for the learning task $\mathcal{T}_t$ to improve the performance of the prediction function $f_{\mathcal{T}}(\cdot)$ (non-linear function of deep neural networks) [9]. Network-based DTL, one popular implementation method of the DTL categories, refers to the use of a pre-trained network model in the original domain and transforms it into a deep neural network for the target domain by modifying part of its network structure and connection parameters. Fine-tuning described earlier is one way to implement this approach. In general, fine-tuning includes two aspects: to fine-tune the whole network and to fine-tune some certain layers [10]. The only difference is the second method will freeze certain layers with fine-tuning the weights of other layers. The most commonly adopted pre-training models are the several proved successful network structures such as AlexNet, Vgg19 and ResNet. Through the fine-tuning, the advantages of existing model are evident:

- Simplified implementation: No need to build the deep network from new.
- Time-saver: Save the time cost of retraining the model.
- Superior performance: Apply the pre-trained network with more robust and better generalization ability.

III. Experiment

A. Dataset

The dataset used in the training and testing phase is the public data set III from BCI Competition II, which is a well-established dataset for the task of MI classification [11]. Firstly, three bipolar EEG channels (C3, CZ, and C4) were connected to the brain of a 25-year-old healthy female subject shown in Figure 2. When the system gives a random prompt, the subject would move a feedback bar by imagining using the left or right hand. The EEG signals of a total of 9 seconds before and after the motor imagery were collected and sampled with 128 Hz as a subset of the data set. Hence, each trial would have 1152 sampling points. A total of 140 trials were collected in the end. Because there was a low correlation between the signal on the CZ channel and MI classification, only two channels of C3 and C4 were used in the next training process.

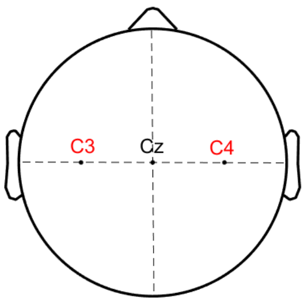

Fig. 2. Locations of Two Utilized EEG Channels (C3 and C4)

B. Data Preprocessing

After loading the dataset, there were two variables x_{train} of $1152 \times 3 \times 140$ storing the EEG signals and y_{train} of 140×1 containing the class-labels '1', '2' for left and right. The FIR band-pass filter is set to only retain the effective EEG band of 8-30 Hz to remove the non-physiological artifacts. Moreover, the wavelet transform threshold denoising method is adopted to eliminate the EOG artifacts. The wavelet function adopted Daubechies 4 (db4) which is very suitable for EEG signal processing. Through multiple tests and comparisons, the number of decomposition levels was finally set to 2. While denoising, the most of the vital characteristics of the EEG signal could be preserved. Then a soft threshold function was applied so that the overall continuity of the wavelet coefficients obtained by the processing would be better, the signal after denoising would be smoother, and the oscillation after denoising would be avoided. The selection form of the threshold adopted unbiased likelihood estimation (rigrsure rule), which is an adaptive threshold selection based on Stein's unbiased likelihood estimation principle. Besides, the threshold used was set to adjust the threshold according to different noise estimates to better preserve the local characteristics of the signal. The comparison graph between the original signals and de-noised signals is displayed in Figure 3.

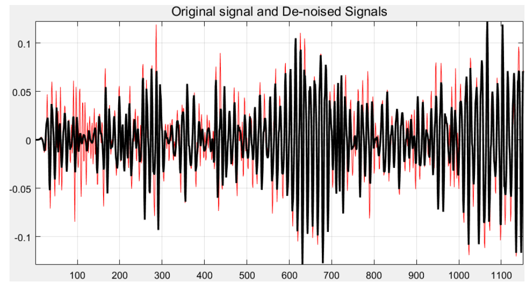

Fig. 3. Original Signals (Red) and De-noised Signals (Black)

C. Feature Extraction

The DTL model used next is the pre-trained AlexNet architecture. The input of model is two-dimensional image data instead of one-dimensional EEG signals. Therefore, a continuous wavelet transform is used to convert the EEG signal into an equivalent image containing time, scale, and amplitude information. The analytic Morlet wavelet was finally adopted as the wavelet base, because it can maintain time and frequency with equal variance for time-frequency analysis. Based on the EEG signals collected through two channels simultaneously, the project pieced two channels into one array to generate one image for each trial displayed in Figure 4. This would be handy as input for CNN model training.

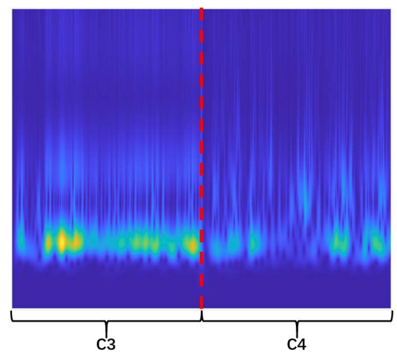

Fig. 4. Generated Image with CWT (C3 on the Left and C4 on the Right)

According to the input requirements of AlexNet, the size of the generated two-dimensional image is rescaled to 227×227 pixel. By using the time amplitude values of different scales as the features of motor imagination classification, the AlexNet model is used for training and testing. After all the 140 trials of data were generated and converted into images of 227×227 pixels, these images were stored into two folders according to the corresponding labels ('1' is for left and '2' is for right).

D. Network Training

Originally, the pre-trained AlexNet framework had 1000 classification results, so the final layers needed to be fine-tuned to suit for the two classification results shown in Figure 5. Last three fully connected layers of the framework were removed and replaced by one fully connected layer, one softmax layer, and one classification layer with the two classification results as the target. The images previously stored in the folder were randomly divided into training data sets and test data sets at a ratio of 8:2. As a contrast, this paper established another comparison group to test the classification performance of non-DTL AlextNet by using 10% as the training set and 90% of the data as the test set. With many times of debugging and comparison, the hyper-parameters were set: firstly, the stochastic gradient descent method was applied, which is very common in the CNN framework utilizing a single training data to update the model parameters for greatly accelerating the training speed. Then the mini-batch size, epoch maximum and initial learning rate were set to 7, 18 and 0.00008.

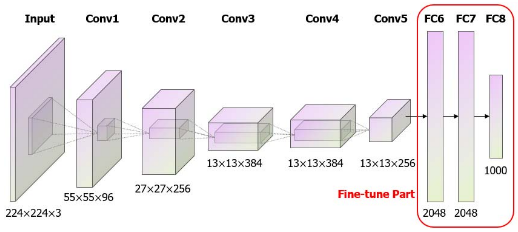

Fig. 5. Model Framework of AlexNet After DTL Method

IV. Results and analysis

The most important measurement of the efficiency for MI classification algorithm is the accuracy directly, and the calculation equation is shown in Equation 3 (TP is truly positive, TN is truly negative, FP is false positive and FN is false negative):

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{3}$$

The specific results of the training model using DTL were shown in Figure 7. The accuracy of the validation rate is achieved at 96.43%. For the comparison group whose training set and test set are 1:9 representing the DL without TL, the final results of testing displayed was just 53.97%. Additionally, this paper calculated the Kappa coefficient which is responsible for the consistency testing of accuracy evaluation. The specific equations are shown in Equation 4 and 5 which includes the actual accuracy ($Accuracy$), random accuracy ($R_{Accuracy}$), and the number of total classes (N).

$$Kappa = \frac{Accuracy - R_{Accuracy}}{1 - R_{Accuracy}} \tag{4}$$

$$R_{Accuracy} = \frac{1}{N} \tag{5}$$

The value of Kappa coefficient is calculated between -1 and 1 but is usually between 0 and 1. Depending on the different values of the Kappa, the model can be divided into five groups to distinguish between different levels of consistency shown in Table I.

TABLE I
Different Levels of Consistency by Different Value of Kappa.

Ranges	0-0.2	0.21-0.4	0.41-0.6	0.61-0.8	0.81-1
Levels	slight	fair	moderate	substantial	almost exact

After the calculation, the values of the Kappa coefficient are 0.9286 for the DTL model and 0.00794 for the DL method without TL. Hence, the results of DTL are almost exactly consistent. Under this comparison of the reference parameter, the improvement of classification prediction by applying the DTL method is more and more evident. Moreover, compared with other related research results that use the same data set but different machine learning methods, the results of the classification model based on DTL still had obvious advantages. These previous research results are arranged and summarized in Table II and Figure 6. This again demonstrates the powerful performance of the CWT-DTL method and the value of the research direction of this project. Therefore, this paper concludes that the application of the CWT-DTL can significantly improve the training of MI classification models.

TABLE II
Related Research Results using the Same Dataset.

Years	Researchers	Models	Accuracies
2015	Masoomi [12]	LDA-based wrapper SFS	90%
2016	Jang [13]	STFT with kNN	83.57%
2016	Das [14]	WT+SE using SVM and kNN	86.4%
2016	Bashar [15]	MEMD + STFT with kNN	90.71%
2017	Chatterjee [16]	Fuzzified Adaptation with SVM	81.48%
2017	Mirvaziri [17]	RTPSO with FFNN	81%
2019	Eslahi [18]	Genetic Algorithm with FKNN	84%
2019	Shovon [19]	STFT with CNN	89.73%
2019	Lee [20]	CWT with 1D CNN	92.9%
2020	Kant [21]	WPT+CWT with CNN	95.71%
2021	This paper	WTTD + CWT with CNN	96.43%

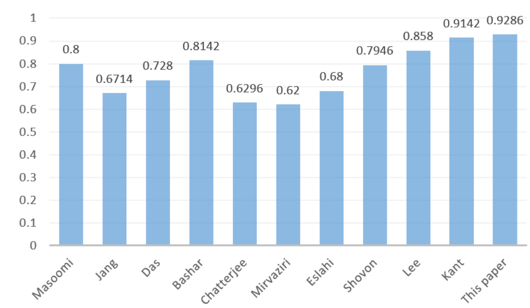

Fig. 6. Kappa Values for Related Research using the Same Dataset

V. Conclusion

A classification method for two types of MI-EEG signals separating left-hand and right-hand motor imaging tasks is proposed in this paper. Applying BCI Competition II Data Set III, 140 trials were extracted from 2 EEG channels as training and test sets. The method of CWT is adopted to convert one-dimensional signals into two-dimensional images and then used transfer learning to train a pre-trained CNN to classify the generated images. The final classification accuracy and Kappa coefficient using DTL reach 96.43% and 0.9286. By comparing the results of the control group of the DL method without TL and other researchers on the same data set with different machine learning methods, the results of the study offered preferably empirical performance.

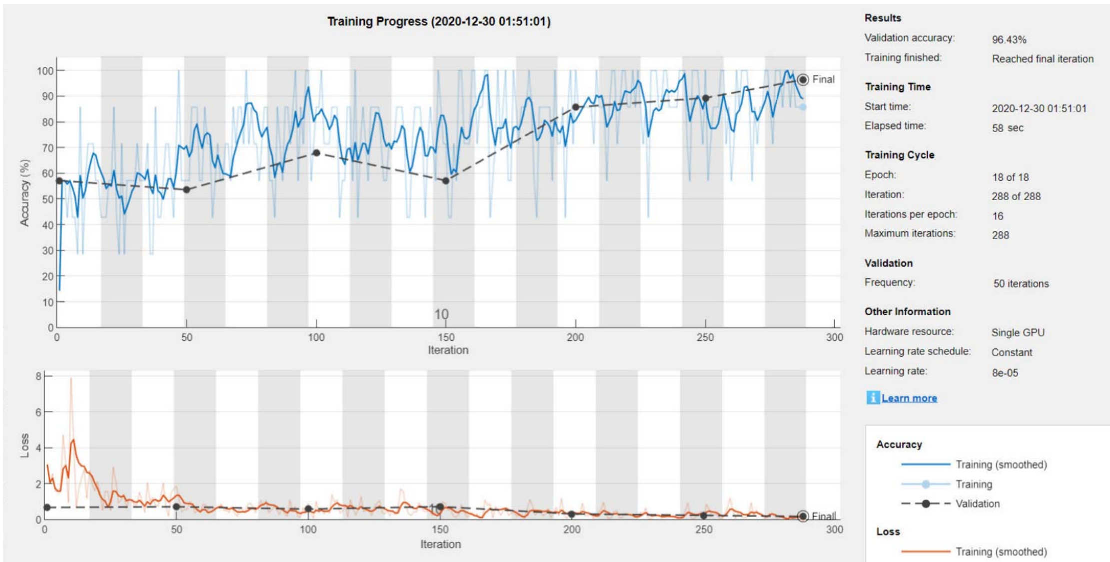

Fig. 7. Performance Evaluation With the DTL Methods

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